



Week 2026-W15(04/06-04/12)

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Weekly Progress Overview

Finished

- Procured necessary sensors and verified hardware compatibility
- Completed hardware assembly for the multi-channel signal collector
- Drafted the initial SVM.py scripty for classification

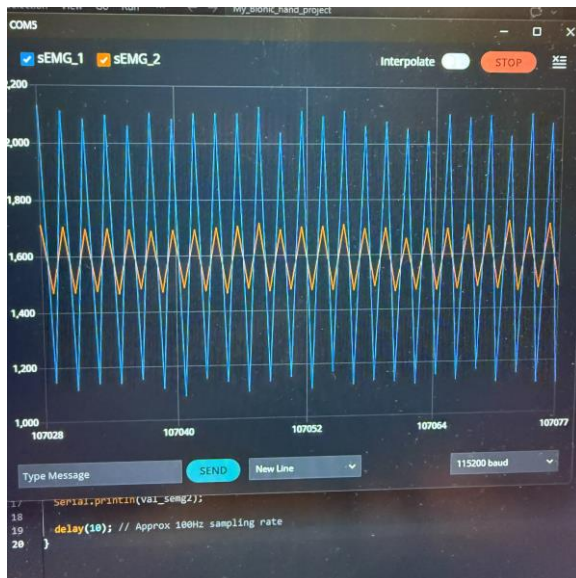
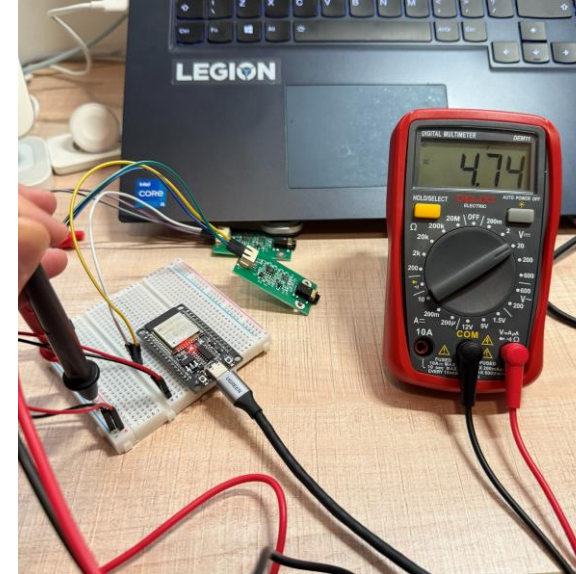
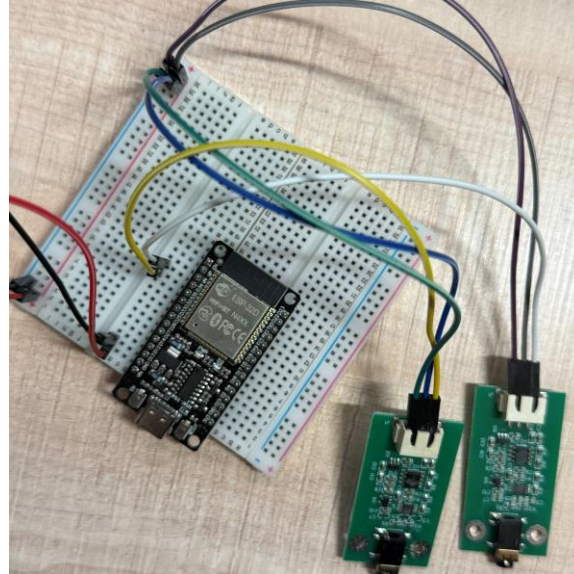
Postponed

- Update the data collection script to add gesture labels

Demonstrations



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- Hardware setup completed
- Able to detect muscle signal for both inner and outer side of forearm



Demonstrations

	A	B	C	D	E	F	G	H
1	Timestamp	Inner_Env	Outer_Env	Index_Dist	Mid_Dist	Ring_Dist	Pinky_Dist	
2	1.78E+09	0	0	0	0	0	0	
3	1.78E+09	487.9	582.44	0	0	0	0	
4	1.78E+09	487.2	582.52	0	0	0	0	
5	1.78E+09	487.8	583	0	0	0	0	
6	1.78E+09	487.74	582.96	0	0	0	0	
7	1.78E+09	485.84	583.46	0	0	0	0	
8	1.78E+09	485.54	583.16	0	0	0	0	
9	1.78E+09	485.44	583.82	0	0	0	0	
10	1.78E+09	485.24	583.56	0	0	0	0	
11	1.78E+09	486.02	584.2	0	0	0	0	
12	1.78E+09	485.94	584.06	0	0	0	0	
13	1.78E+09	484.6	584.36	0	0	0	0	
14	1.78E+09	484.4	584.06	0	0	0	0	
15	1.78E+09	482.42	584.1	0	0	0	0	
16	1.78E+09	481.88	584.16	0	0	0	0	
17	1.78E+09	481.1	584.42	0	0	0	0	
18	1.78E+09	481.08	584.42	0	0	0	0	
19	1.78E+09	479.2	584.3	0	0	0	0	
20	1.78E+09	477.04	584.44	0	0	0	0	
21	1.78E+09	476.6	584.42	0	0	0	0	
22	1.78E+09	476.18	584.62	0	0	0	0	
23	1.78E+09	473.42	584.52	0	0	0	0	
24	1.78E+09	472.42	584.52	0	0	0	0	
25	1.78E+09	472.84	584.7	0	0	0	0	
26	1.78E+09	472.08	584.74	0	0	0	0	
27	1.78E+09	471.44	585.26	0	0	0	0	
28	1.78E+09	471.68	585.24	0	0	0	0	

- Developed sync_collector.py for real-time data acquisition
- Achieved temporal alignment between serial sEMG data and visual distance measurements

Next:

Write a new script to implement tagging (gestures) for transition from raw data to labelled data.



Blockers / Questions

- Literature Review: Current theoretical understanding of sEMG control requires deepening; this will be an ongoing background focus.
- Baseline Model Request: Before developing custom optimisations, it is prudent to replicate a mature, open-source sEMG project.
- Discussion point: Could the supervisor recommend a reliable open-source repository as a starting baseline?



Next Week Preview

- Integrate gesture labelling logic into the preprocessing pipeline.
- Explore feasible methods for data visualisation based on initial results.
- Conduct preliminary data collection to evaluate system accuracy and signal reliability.